

- 7 Annie writes a paragraph of text as an answer to an examination question about programming languages.

Using the list given, complete Annie's answer by inserting the correct **six** missing terms. Not all terms will be used.

- Assembly
- Converter
- Denary
- Hexadecimal
- High-level language
- Low-level language
- Machine Code
- Source Code
- Syntax
- Translator

The structure of language statements in a computer program is called the

..... . A programming language that uses natural language statements is called a When programs are written in this type of language they need a to convert them into

A programming language that is written using mnemonic codes is called a

..... . An example of this type of language is language.

[6]

(c) The computer has an operating system.

(i) A signal causes the operating system to stop and assess what to do next.

Identify the name of this signal.

..... [1]

(ii) State **two** functions of an operating system.

1

2

[2]

6 Ishan is a member of a software community that develops computer games. He has programmed a new feature for one of the community's existing games.

(a) Ishan compiles the program before he issues it to the community.

(i) Explain **one** benefit of Ishan compiling the program.

.....
..... [1]

(ii) Explain **one** drawback of Ishan compiling the program.

.....
..... [1]

2 A programmer uses a high-level language to write a computer program.

(a) Four statements are given about high-level programming languages.

Tick (✓) to show if each statement is **True** or **False**.

Statement	True (✓)	False (✓)
High-level languages need to be translated into machine code to run on a computer		
High-level languages are written using mnemonic codes		
High-level languages are specific to the computer's hardware		
High-level languages are portable languages		

[4]

(b) Tick (✓) to show which of the following is an example of a high-level language program.

Example program	Tick (✓)
1011100000110000 0000011011100010	
INP STA ONE INP STA TWO ADD ONE	
a = input() b = input() if a == b: print("Correct") else: print("Incorrect")	

[1]

(b) Computer systems often use interrupts.

Five statements are given about interrupts.

Tick (✓) to show if each statement is **True** or **False**.

Statement	True (✓)	False (✓)
Interrupts can be hardware based or software based		
Interrupts are handled by the operating system		
Interrupts allow a computer to multitask		
Interrupts work out which program to give priority to		
Interrupts are vital to a computer and it cannot function without them		

[5]

- 4 Assemblers, compilers and interpreters are types of translators.

Tick (✓) to show which statements apply to each translator. Each statement may apply to more than one type of translator.

Statement	Assembler (✓)	Compiler (✓)	Interpreter (✓)
Translates low-level language to machine code			
Translates high-level language to machine code			
Produces error messages			
Translates high-level language one line at a time			
Produces an executable file			

[5]

- 2 Both an interpreter and a compiler can be used when writing a program in a high-level language.

(a) Explain why a programmer would make use of both an interpreter **and** a compiler.

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..... [4]

(b) Give **three** reasons why a programmer would choose to write a program in a high-level language, instead of a low-level language.

Reason 1

.....

Reason 2

.....

Reason 3

.....

[3]

(c) An interrupt signal is sent from the printer to the computer.

(i) Give **two** examples of when a printer would generate an interrupt signal.

Example 1

Example 2

[2]

(ii) Many devices send interrupt signals.

Identify the software in the computer that will receive and manage all interrupt signals.

..... [1]

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7 Adeel has used a high-level language to program a mobile application.

(a) Describe what is meant by a high-level language.

.....
.....
.....
..... [2]

(b) Adeel uses an interpreter while developing and testing the application.

Adeel uses a compiler when the application is ready to be shared with others.

Compare the features of interpreters and compilers.

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..... [4]

(c) Adeel is considering distributing his application as free software or shareware.

Explain the difference between free software and shareware.

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.....

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..... [5]

- -

(c) Computers based on the Von Neumann model, for a computer system, use interrupts.

Explain why interrupts are needed.

.....

.....

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..... [2]

- 5 Six statements are given about programming languages.

Tick (✓) to show whether each statement applies to high-level language, assembly language or machine code. Some statements may apply to more than one type of programming language.

Statement	High-level language (✓)	Assembly language (✓)	Machine code (✓)
it requires a translator to be processed by a computer			
it is an example of low-level language			
it uses mnemonics			
it uses English-like statements			
it can be used to directly manipulate hardware in the computer			
it is portable			

[6]

- 5 Tamaz stores confidential data on his computer.

He uses the Internet regularly and is concerned about his data being viewed by unauthorised people. He currently has **one** software method to stop his data being viewed, which is a password.

He wants to add other software methods to stop his data being viewed by unauthorised people.

- (a) State **two** other software methods that Tamaz could use to stop his data being viewed by unauthorised people.

1

2 [2]

- (b) Tamaz's computer has an operating system. **Two** functions of the operating system are file management and memory management.

State **two** other functions of the operating system.

1

2 [2]

- 8 Victoria develops a computer game to sell on a gaming website. She writes her program using English-like statements.

- (a) State which type of programming language Victoria is using.

..... [1]

- (b) Victoria uses **two** different types of translator when creating the program for the computer game.

State which translator is the most suitable for the given tasks.

Give the benefits of using that translator for the task.

You must choose a different translator for each task.

- (i) To translate the code during development of the game.

Translator

Benefits

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..... [3]

- (ii) To translate the final program and upload to the website for distribution, without the source code.

Translator

Benefits

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[3]

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- 6 A programmer can use translators, such as an interpreter and a compiler, when developing a computer program.

- (a) Give **one** similarity between a compiler and an interpreter.

.....

..... [1]

- (b) Describe **two** differences between a compiler and an interpreter.

Difference 1

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Difference 2

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.....

[4]

- (c) Identify **one** other type of translator.

..... [1]

9 Padma opens an application on her computer.

An interrupt is generated to inform the Central Processing Unit (CPU) that the application has been opened.

(a) Give **three** other examples of when an interrupt signal could be generated.

1

2

3 [3]

(b) State what would happen if interrupt signals were **not** used in a computer.

.....

..... [1]

- 4 Frederick prints a document that he has typed.

The printer begins to print the document, but then a message is displayed on Frederick's computer to say that the paper has jammed.

- (a) Describe the role of an interrupt in generating a message on the computer that the paper has jammed.

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..... [4]

- (b) Give **two** other examples of when an interrupt signal could be generated.

1

2 [2]

- 7 Julius creates a computer application that calculates how many years it will take to pay for a house.

- (a) Julius uploads his application to his website for people to download. Before he uploads the application, he translates the code using a compiler.

Explain why Julius uses a compiler, rather than an interpreter, to do this.

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..... [4]

7 Ishani is a software developer who is creating a new computer game.

(a) Ishani uses an interpreter and a compiler at different stages of the game creation.

(i) Explain when it is most appropriate for Ishani to use an interpreter.

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..... [3]

(ii) Explain when it is most appropriate for Ishani to use a compiler.

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..... [3]

6 A compiler and an interpreter are two different types of translator.

(a) One similarity between a compiler and an interpreter is that they both translate high-level language into machine code.

(i) Give **one** other similarity between a compiler and an interpreter.

.....
..... [1]

(ii) Explain **two** differences between a compiler and an interpreter.

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..... [4]

4 Jason is a programmer who writes computer programs in a high-level language.

(a) Describe what is meant by a high-level language.

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..... [3]

8 The paragraph provides information about translators.

Complete the paragraph using the most appropriate terms from the list. **Not** all terms in the list need to be used.

- all at once
- assembler
- assembly language
- binary
- commenting
- compiling
- debugging
- executable file
- high-level language
- interpreted
- is **not** required
- is required
- line by line
- low-level language

A compiler and an interpreter translate into machine code. An interpreter does this by translating and executing the code and a compiler does this by translating and executing the code

One benefit of a compiler is that it creates an, which means that a translator each time to run the file. is easier to do with an interpreter because it stops translating and reports an error at the place where it occurs in the code.

- 2 A programmer has designed a three-dimensional (3D) interactive computer game. They are going to develop a program for the game. The program needs to run efficiently, but it must also be developed as soon as possible.
- (a) Tick (✓) **one** box to identify whether the programmer should use a high-level language or a low-level language to develop the program.

Explain the reasons for your choice.

High-level language

Low-level language

11

Reasons for your choice

[4]

- (b)** If the programmer chooses a high-level language, they can use a compiler or an interpreter to translate the high-level language into a low-level language.

Describe the operation of a compiler and of an interpreter.

Compiler

.....

.....

.....

Interpreter

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.....

[4]

- 4 Complete the statements about different types of software.

Use the terms from the list.

Some of the terms in the list will **not** be used. You should only use a term once.

application assembly language bootloader central processing unit (CPU)
firmware hardware operating output system user

..... software provides the services that the computer requires; an example is utility software.

..... software is run on the operating system.

The system is run on the firmware, which is run on the

[4]

- 7 A programmer uses a low-level language to write a computer program for a vending machine.

(a) Describe what is meant by a low-level language.

.....
.....
.....
..... [2]

(b) Give **two** reasons why the programmer would choose to write the computer program in a low-level language instead of a high-level language.

1
.....
2
.....

[2]

5 A programmer writes a computer program using a high-level language.

(a) Tick (✓) **one** box to show which statement is correct about writing computer programs in a high-level language.

A Mnemonics are used to create instructions.

☐

B The computer program is harder to debug than a low-level language program.

☐

C The computer program is machine independent.

☐

D The hardware of the computer can be directly manipulated.

☐

[1]

(b) The programmer uses a compiler to translate the computer program.

(i) Describe how the compiler translates the computer program.

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..... [3]

(ii) Describe how the compiler reports errors.

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..... [2]

(c) The programmer uses an integrated development environment (IDE) to create the computer program.

One function of the IDE is that it has the built-in compiler.

Give **three** other common functions of an IDE.

1

2

3

[3]

10 A user has both system software and application software installed on their computer.

(a) Describe the difference between system software and application software.

Give an example of each software in your answer.

.....

.....

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..... [4]

(b) State which component in the computer would store both types of software when the power is turned off.

..... [1]



4 A programmer writes a computer program in a high-level language.

(a) Tick (✓) **one** box to show which statement is a benefit of writing a program in a high-level language, instead of a low-level language.

A The program can directly manipulate the hardware.

☐

B The program is machine independent.

☐

C The program is more memory efficient.

☐

D The program is quicker to execute.

☐

[1]

(b) Translators are used to translate the high-level language so that it can be processed by the computer.

(i) State what the high-level language is translated into.

..... [1]

(ii) One translator converts and executes the code line by line.

Identify which type of translator would do this.

..... [1]

(iii) One translator creates an error report displaying all the errors in the code before it can be executed.

Identify which type of translator would do this.

..... [1]

(iv) One translator creates an executable file.

Identify which type of translator would do this.

..... [1]

11 Software is installed on a computer to manage files, memory and multitasking.

(a) State the name of the software that can do these tasks.

..... [1]

(b) Give **one** task that the software allows the user to do to manage files.

..... [1]

(c) Describe what is meant by managing memory.

.....
.....
.....
..... [2]

(d) A signal is sent within the computer to allow multitasking to occur.

State the name of this type of signal.

..... [1]

- 3 The table contains **four** descriptions about a computer system.

Complete the table by writing the correct term for each description.

Term	Description
.....	A collective term for the physical components of the computer system.
.....	A type of software that provides services that the user requires and allows the user to perform tasks on the computer.
.....	A type of software that manages the main functions of the computer, including managing files and managing memory.
.....	A type of software that is stored in the read only memory (ROM). It includes the basic input output system (BIOS) and the bootloader.

[4]

9 An interrupt is a type of signal that is used in a computer.

(a) State the name of the type of software that manages interrupts.

..... [1]

(b) Describe how interrupts are used when a key is pressed on a keyboard.

[5]

(c) Interrupts can be hardware based or software based.

A key press is one example of a hardware interrupt.

(i) Give **two** other examples of a hardware interrupt.

1

2

[2]

(ii) Give **two** examples of a software interrupt.

1

.....

2

.....

[2

(d) The mobile telephone has an operating system.

Describe the purpose of the operating system.

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..... [3]